

# Deep Learning Notation Cheat Sheet

## Main Idea

In deep learning, symbols are not universal. Different authors may use different symbols for the same concept. For example, model parameters may be written as

$$\theta, \quad w, \quad W, \quad \phi, \quad \psi, \quad \omega.$$

Therefore, the most important habit is not memorizing symbols, but asking:

What object does this symbol represent?

## Common Deep Learning Symbols

| Symbol                      | Often Means                             | Plain English Meaning                                   | Common Alternatives            |
|-----------------------------|-----------------------------------------|---------------------------------------------------------|--------------------------------|
| $\theta$                    | Model parameters                        | All learnable parameters of the model                   | $w, W, \phi, \psi, \omega$     |
| $w$                         | Weight                                  | A learnable scalar parameter                            | $\theta, \beta$                |
| $W$                         | Weight matrix                           | A matrix of learnable weights                           | $\Theta, A$                    |
| $b$                         | Bias                                    | Learnable offset term                                   | $c, \beta_0$                   |
| $\phi$                      | Encoder or feature extractor parameters | Parameters of one part of the model                     | $\theta_e, \psi$               |
| $\psi$                      | Auxiliary model parameters              | Often used for critic, value network, or another module | $\theta_v, \omega$             |
| $\omega$                    | Generic parameters                      | Another symbol for learnable parameters                 | $\theta, w$                    |
| $x$                         | Input                                   | Data fed into the model                                 | $X, \mathbf{x}$                |
| $y$                         | Target label                            | Correct answer or ground truth                          | $t, r$                         |
| $\hat{y}$                   | Prediction                              | Model output estimate                                   | $y_{\text{pred}}, \tilde{y}$   |
| $f_\theta$                  | Model / neural network                  | A function parameterized by $\theta$                    | $h_\theta, g_\theta, M_\theta$ |
| $p_\theta(y   x)$           | Probabilistic model                     | Model's predicted probability distribution              | $P(y   x; \theta)$             |
| $\mathcal{L}$               | Loss function                           | Measures how bad the prediction is                      | $L, J, \text{cost}$            |
| $J(\theta)$                 | Objective function                      | Function to minimize or maximize                        | $\mathcal{L}(\theta)$          |
| $\nabla_\theta \mathcal{L}$ | Gradient                                | Direction showing how the loss changes with parameters  | grad                           |
| $\eta$                      | Learning rate                           | Step size during training                               | $\alpha, \text{lr}$            |
| $\alpha$                    | Learning rate or coefficient            | Often a step size or weighting factor                   | $\eta, \lambda$                |

| Symbol               | Often Means                        | Plain English Meaning                                    | Common Alternatives      |
|----------------------|------------------------------------|----------------------------------------------------------|--------------------------|
| $\lambda$            | Regularization strength            | Controls penalty term                                    | weight decay coefficient |
| $\beta$              | Momentum coefficient or parameter  | Often used in optimization or statistics                 | decay rate               |
| $z$                  | Logit or latent variable           | Raw score before activation, or hidden latent factor     | latent code              |
| $h$                  | Hidden representation              | Internal feature vector                                  | embedding, hidden state  |
| $a$                  | Activation or action               | In neural nets: activation; in RL: action                | $h$                      |
| $\sigma$             | Activation function                | Nonlinear function                                       | sigmoid, activation      |
| $D$ or $\mathcal{D}$ | Dataset                            | Collection of training examples                          | training set             |
| $n$ or $N$           | Number of samples                  | Dataset size or sample count                             | $m$                      |
| $B$                  | Batch size                         | Number of examples used per update                       | mini-batch size          |
| $i$                  | Index                              | Refers to the $i$ -th example                            | sample index             |
| $t$                  | Time step or target                | In sequence models: time; in supervised learning: target | step                     |
| $T$                  | Sequence length                    | Total number of time steps                               | horizon                  |
| $d$                  | Dimension                          | Size of input or feature vector                          | feature dimension        |
| $K$                  | Number of classes                  | Total class count                                        | $C$                      |
| $C$                  | Classes or channels                | In classification: classes; in CNNs: channels            | $K$                      |
| $\mathbb{R}^d$       | Real vector space                  | A vector with $d$ real-valued entries                    | shape notation           |
| $\mathbb{E}$         | Expectation                        | Average value under a distribution                       | mean                     |
| $\sim$               | Sampled from                       | Indicates a random variable follows a distribution       | drawn from               |
| $\propto$            | Proportional to                    | Equal up to a constant factor                            | proportional             |
| $\arg \max$          | Argument of maximum                | Choose the input giving the largest value                | best choice              |
| $\arg \min$          | Argument of minimum                | Choose the input giving the smallest value               | optimizer target         |
| $\theta^*$           | Optimal parameters                 | Best parameters after training                           | $\hat{\theta}$           |
| $\hat{\theta}$       | Estimated parameters               | Parameters estimated from data                           | $\theta^*$               |
| $q_\phi$             | Approximate distribution / encoder | Common in variational inference and VAEs                 | inference model          |
| $p_\theta$           | Generative model / decoder         | Model distribution parameterized by $\theta$             | likelihood model         |
| $\pi$                | Policy or probability distribution | In RL, the model that chooses actions                    | $p, q, \mu$              |
| $s$                  | State                              | Situation in reinforcement learning                      | observation              |
| $a$                  | Action                             | Action taken by an agent                                 | control                  |
| $r$                  | Reward                             | Feedback signal in reinforcement learning                | reward value             |
| $Q(s, a)$            | Action-value function              | Expected return after taking action $a$ in state $s$     | critic                   |
| $V(s)$               | Value function                     | Expected return from state $s$                           | state value              |

| Symbol                     | Often Means                 | Plain English Meaning                                      | Common Alternatives |
|----------------------------|-----------------------------|------------------------------------------------------------|---------------------|
| $\gamma$                   | Discount factor             | Controls importance of future rewards                      | discount            |
| $\tau$                     | Trajectory or temperature   | In RL: sequence of states/actions; in softmax: temperature | path, temp          |
| $\mathcal{N}(\mu, \Sigma)$ | Normal distribution         | Gaussian distribution with mean and covariance             | Gaussian            |
| $\mu$                      | Mean                        | Average or center of a distribution                        | expectation         |
| $\Sigma$                   | Covariance matrix           | Spread and correlation of variables                        | covariance          |
| $\epsilon$                 | Small constant or noise     | Used for numerical stability or randomness                 | noise               |
| $D_{\text{KL}}$            | KL divergence               | Difference between two distributions                       | relative entropy    |
| CE                         | Cross entropy               | Common classification loss                                 | log loss            |
| MSE                        | Mean squared error          | Common regression loss                                     | $\ell_2$ loss       |
| NLL                        | Negative log-likelihood     | Loss based on probability model                            | log loss            |
| SGD                        | Stochastic gradient descent | Basic gradient-based optimizer                             | optimizer           |
| Adam                       | Adaptive optimizer          | Popular optimizer using moving averages                    | optimizer           |

## Very Common Patterns

### Model Prediction

$$\hat{y} = f_{\theta}(x)$$

This means:

$$\text{Input } x \longrightarrow \text{model } f_{\theta} \longrightarrow \text{prediction } \hat{y}.$$

Here,  $\theta$  means the learnable parameters of the model.

### Training Objective

$$\theta^* = \arg \min_{\theta} \mathcal{L}(\theta)$$

This means:

Find the best parameters  $\theta^*$  that minimize the loss  $\mathcal{L}(\theta)$ .

### Gradient Descent Update

$$\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}(\theta)$$

This means:

new parameters = old parameters – learning rate  $\times$  gradient.

## Encoder and Classifier Example

$$z = f_\phi(x), \quad \hat{y} = g_\theta(z)$$

This usually means:

$f_\phi$  = encoder with parameters  $\phi$ ,

$g_\theta$  = classifier or decoder with parameters  $\theta$ .

So  $\phi$  and  $\theta$  are both parameters, but they belong to different parts of the model.

## How to Decode Symbols in a Paper

When reading a paper, ask the following questions:

| Question                                 | Common Symbols                         |
|------------------------------------------|----------------------------------------|
| Is this data?                            | $x, y, \mathcal{D}, x_i, y_i$          |
| Is this a learnable parameter?           | $\theta, w, W, b, \phi, \psi, \omega$  |
| Is this a model or function?             | $f, g, h, p_\theta, q_\phi$            |
| Is this a loss or objective?             | $L, \mathcal{L}, J$                    |
| Is this a hyperparameter?                | $\eta, \alpha, \lambda, \beta, \gamma$ |
| Is this a probability distribution?      | $p, q, \pi, \mathcal{N}$               |
| Is this reinforcement learning notation? | $s, a, r, \pi, Q, V, \gamma$           |

## Key Takeaway

The most important rule is:

$\theta, w, W, \phi, \psi, \omega$  often all mean learnable parameters.

Authors use different letters mainly to separate different parts of the model.

For example:

$$p_\theta(x \mid z), \quad q_\phi(z \mid x)$$

usually means:

$p_\theta$  = decoder or generative model,

$q_\phi$  = encoder or approximate inference model.

Both  $\theta$  and  $\phi$  are learnable parameters, but they belong to different networks.